



NCIE-015

# DevSecOps, Infrastructure & Deployment Specification

**In development — for human review of full expansion —  
Version 1.0**

Operationalizes the approved NCIE-001–014 foundation: environments/promotion, infrastructure-as-code, source control, build/artifact supply chain, CI gates, release management, deployment strategies, runtime/Agent-sandbox/network operations, IAM/secrets operations, database/storage and integration operations, ARGUS/model/functional-module/frontend delivery operations, observability, SLOs, alerting/on-call/incident management, runbooks, change/maintenance, capacity, backup/DR, vulnerability management, security operations, operational audit/provenance, reconciliation/drift, go-live readiness, and DevSecOps/operations testing.

Operational-only, upstream-consuming: NCIE-015 owns deployment/infrastructure realization, CI/CD operationalization, runtime operations, observability and incident/runbook execution; it does not redefine product, canonical data, ARGUS, Memory, Agent/VPF, governance, IAM, Evidence, API, UX, domain-module or persistence semantics.

Expands approved NCIE-015 Skeletal v0.1, itself traced to approved NCIE-001 v1.3 through approved NCIE-014 v1.0.

Prepared: 16 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

# Document Governance

## 0.1 Formal-Title Reconciliation (Directive Item 3)

The approved NCIE-015 Skeletal v0.1 carries the source-supported working title 'NCIE-015 — DevSecOps, Infrastructure & Operations Specification' and its own Title-control note directs that the exact formal wording be reconciled against the controlled suite register before full expansion. The authoritative source located and used is the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), Section 3, whose NCIE-015 entry records the formal title as 'DevSecOps, Infrastructure & Deployment Specification' with Primary Purpose 'Environments, infrastructure, CI/CD, observability, resilience and deployment' — matching this document's actual 36-chapter operational scope (environments, IaC/CI/CD, runtime/network/storage/integration operations, observability, incident/recovery, capacity, security operations) exactly. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register's final word is 'Deployment' where the skeleton's working title used 'Operations'; no material scope discrepancy was found, so this is a wording correction only, per directive item 3, not a Human Review halt). Per this result, the document-control formal title has been set to 'NCIE-015 — DevSecOps, Infrastructure & Deployment Specification' throughout this document (cover, this section, Document Governance table, Authority Chain, running header, Chapter 36 disposition statement, core document properties and the deliverable filename); descriptive prose referring to the document's 'operations' content throughout the chapters is unaffected, since that describes subject matter, not the formal document-control title. This determination is recorded as Human Review item HR15-1-1 for institutional confirmation rather than treated as silently final.

## 0.2 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification → NCIE-012 UX/UI & Design System Specification → NCIE-013 Functional Modules Specification → NCIE-014 Database & Storage Engineering Specification → NCIE-015 DevSecOps, Infrastructure & Deployment Specification. NCIE-015 operationalizes governed architecture, security, data, domain and persistence decisions; it creates no canonical meaning, institutional authority, security policy, Evidence semantics, API contract, UX pattern, domain-business rule or persistence semantic of its own.

## 0.3 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–014 authority the chapter operationalizes, stating only the operational relationship rather than restating upstream architectural, security, canonical, domain-business or persistence content.

## 0.4 Governing DevSecOps / Operations Invariants Register

These 18 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-015 Skeletal v0.1 and by the approved NCIE-001–014 authority chain; no institutional decision is required to adopt them. The upstream canonical, governance, security, Evidence/provenance, integration, UX, domain-module and persistence invariants already binding from NCIE-003 and NCIE-008–014 remain fully in force and are cross-referenced, not restated, in the chapters below.

- DEPLOYED ≠ APPROVED.
- BUILD SUCCESS ≠ RELEASE AUTHORIZATION.
- PIPELINE PASS ≠ PRODUCTION ACCEPTANCE.
- INFRASTRUCTURE ACCESS ≠ CONTENT AUTHORITY.
- OPERATOR PRIVILEGE ≠ INSTITUTIONAL DECISION AUTHORITY.
- CONFIGURATION DEPLOYED ≠ POLICY APPROVED.
- HEALTHY SERVICE ≠ CORRECT BUSINESS OUTCOME.
- TELEMETRY ≠ AUDIT ≠ PROVENANCE.
- ALERT ≠ INCIDENT.

- INCIDENT DECLARED ≠ ROOT CAUSE ESTABLISHED.
- ROLLBACK ≠ HISTORICAL ERASURE.
- FAILOVER ≠ REAUTHORIZATION.
- RESTORE ≠ REAUTHORIZATION.
- AUTOSCALING ≠ CAPACITY ASSURANCE.
- DR EXERCISE ≠ PRODUCTION RECOVERY PROOF.
- RUNBOOK EXECUTION ≠ POLICY EXCEPTION AUTHORITY.
- EMERGENCY CHANGE ≠ PERMANENT POLICY CHANGE.
- AUTOMATION ≠ UNBOUNDED OPERATIONAL AUTHORITY.

## 0.5 Operations/Upstream Boundary Note

NCIE-015 owns deployment/infrastructure realization, CI/CD operationalization, runtime operations, observability, incident/runbook execution, capacity and continuity operations. It consumes NCIE-004 engineering standards, NCIE-008 institutional decision rights, NCIE-009 security/IAM controls, NCIE-010 Evidence/provenance/audit semantics, NCIE-011 integration contracts, NCIE-012 UX delivery needs, NCIE-013 domain-module workloads and NCIE-014 persistence architecture, without redefining any of them. Where a matter requires a platform/product selection or a numeric threshold (SLO, RPO/RTO, alert threshold, capacity limit, remediation deadline, maintenance window, retry count, backup schedule, exercise cadence) not fixed by an approved source, this document states the operational framework and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-015 Skeletal v0.1 authorized this full expansion only; it did not itself approve any platform selection, numeric threshold, on-call model, runbook authority or ownership assignment proposed below.

## 0.6 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 64, a Chapter 35 register item that restates an identical NCIE-008–014 institutional, security, persistence or domain-business question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-015 item is created only for a distinct operations decision. Chapter 35 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §35.4).

## 0.7 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual platform selection, numeric threshold, on-call model or runbook authority proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Nothing in this document may be cited as an approved institutional decision, an implemented pipeline/environment, a validated recovery procedure, or a completed test.

## 0.8 Salient Things to Note Discipline

Directive item 78 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 36 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in every full expansion since. Every chapter below renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-015's lib.py raises a build error if a chapter's salient content is missing), and Chapter 36's mechanical audit (§36.2) re-verifies this against the final document.

## 0.9 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-015 Skeletal v0.1 into the complete 36-chapter specification per the NCIE-015 v0.1→v1.0 Claude Full-Expansion Production Directive: six consolidated matrices, platform/numeric-

invention and release-authority/emergency-authority audits, Human Review Register (Chapter 35) and full traceability/handover (Chapter 36). Formal title reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 at production time (\$0.1) — document-control wording corrected from 'Operations' to 'Deployment'; no material scope discrepancy found.

Authority: Approved NCIE-001 Master PRD v1.3 through approved NCIE-014 Database & Storage Engineering Specification v1.0, and the approved NCIE-015 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 36-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

|                                            |                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Document</b>                            | <b>NCIE-015 DevSecOps, Infrastructure &amp; Deployment Specification</b>                                                                                               |
| <b>Edition</b>                             | In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)                                                                                                |
| <b>Requirements authority</b>              | NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED                                                                                                  |
| <b>Architecture authority</b>              | NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED                                                                                                |
| <b>Canonical data authority</b>            | NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED                                                                                                          |
| <b>Engineering baseline</b>                | NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED                                                                                              |
| <b>ARGUS behavioral authority</b>          | NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED                                                                                            |
| <b>Memory &amp; Context authority</b>      | NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED                                                                                           |
| <b>Orchestration authority</b>             | NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED                                                                                         |
| <b>Governance authority</b>                | NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED                                                                            |
| <b>Security authority</b>                  | NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED                                                                                               |
| <b>Evidence/provenance/audit authority</b> | NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED                                                                                             |
| <b>Integration authority</b>               | NCIE-011 API, Integration & Data Connector Specification, Version 1.0 — APPROVED                                                                                       |
| <b>UX/UI authority</b>                     | NCIE-012 UX/UI & Design System Specification, Version 1.0 — APPROVED                                                                                                   |
| <b>Domain-module authority</b>             | NCIE-013 Functional Modules Specification, Version 1.0 — APPROVED                                                                                                      |
| <b>Persistence authority</b>               | NCIE-014 Database & Storage Engineering Specification, Version 1.0 — APPROVED                                                                                          |
| <b>Prior baseline</b>                      | NCIE-015 DevSecOps, Infrastructure & Operations Specification (working title), Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)                    |
| <b>Status</b>                              | IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture |

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# Mandate, Authority Chain & Operations Boundary

Establish NCIE-015 ownership of DevSecOps/infrastructure/operations within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001–014, especially NCIE-004/009–014.

## 1.0 Salient Things to Note

Operations implements approved architecture; it cannot create authority or redefine semantics — Table 1 fixes this boundary and Table 2 maps every platform domain to its technical operator without inventing organizational offices.

### 1.1 Mandate and Scope

NCIE-015 operationalizes governed architecture, security, data, domain and persistence decisions — it does not create product behavior, canonical meaning, institutional authority, security policy, Evidence semantics, API contracts, UX patterns, domain-business rules or persistence semantics.

### 1.2 Operations Boundary — What NCIE-015 Does Not Own

NCIE-015 does not own product requirements, architecture, canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security enforcement policy, Evidence/provenance meaning, API contracts, UX semantics, domain business rules or persistence semantics. It owns only the operational realization layer above all of them.

Table 1. NCIE-015 operations boundary.

| Concern                                               | Owned By                 |
|-------------------------------------------------------|--------------------------|
| Product requirements/scope                            | NCIE-001                 |
| System architecture                                   | NCIE-002                 |
| Canonical data model                                  | NCIE-003                 |
| Engineering baseline/technology families              | NCIE-004                 |
| ARGUS behavioral authority                            | NCIE-005                 |
| Memory & Context architecture                         | NCIE-006                 |
| AI/Agent/Model orchestration                          | NCIE-007                 |
| Institutional governance, RACI                        | NCIE-008                 |
| Security, IAM, Zero-Trust enforcement                 | NCIE-009                 |
| Evidence lifecycle, provenance, audit                 | NCIE-010                 |
| API/event/connector/integration contracts             | NCIE-011                 |
| UX/UI, presentation, interaction, Design System       | NCIE-012                 |
| Domain business rules, workflows, analytical products | NCIE-013                 |
| Physical database/storage engineering                 | NCIE-014                 |
| Deployment/infrastructure realization, operations     | NCIE-015 (this document) |
| Detailed test execution                               | NCIE-016 (downstream)    |

### 1.3 Operational Ownership Matrix (directive item 59)

The consolidated matrix below maps services/platform domains to technical operator responsibility and upstream governance authority, without inventing organizational offices — actual role holders remain a Human Review assignment.

Table 2. Operational Ownership Matrix.

| Platform Domain                                                                                 | Technical Operator Responsibility                   | Upstream Governance Authority | Owni ng Chap ter |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------|------------------|
| Environments/promotion                                                                          | Environment provisioning, isolation, teardown       | NCIE-004/009/014              | Ch.3             |
| IaC/configuration                                                                               | Module authoring, plan/apply, drift remediation     | NCIE-004/008/009              | Ch.4             |
| Source control/CI/build                                                                         | Repository, pipeline, artifact operation            | NCIE-004/008/009              | Ch.5-7           |
| Release/deployment                                                                              | Release execution, rollout, rollback                | NCIE-008/004                  | Ch.8-9           |
| Runtime/network/Agent sandbox                                                                   | Platform operation, scheduling, isolation           | NCIE-004/007/009              | Ch.10-12         |
| IAM/secrets operations                                                                          | Identity/secret operational delivery                | NCIE-009                      | Ch.13-14         |
| Database/storage operations                                                                     | Operational realization of NCIE-014                 | NCIE-014                      | Ch.15            |
| Integration/core-service/module/frontend operations                                             | Operational realization of NCIE-011/005/007/013/012 | NCIE-005/007/011/012/013      | Ch.16-19         |
| Observability/SLO/alerting/incident                                                             | Platform operation, response                        | NCIE-004/008/009/010          | Ch.20-23         |
| Runbooks/change/capacity/backup/DR/vulnerability/SecOps/audit/rec onciliation/readiness/testing | Procedural and platform operation                   | NCIE-008/009/010/014/016      | Ch.24-34         |

1.R RACI / Control Ownership

Table 3. Chapter 1 RACI / control ownership.

| Function               | Accountable (NCIE-008)      | Responsible (Operations)      | Consulted                       | Informed                       |
|------------------------|-----------------------------|-------------------------------|---------------------------------|--------------------------------|
| Adopt/confirm NCIE-015 | Accountable Authority Class | Operations Architecture Owner | NCIE-004/009-014 chapter owners | All institutional role holders |

|              |                  |                  |
|--------------|------------------|------------------|
| operations   | (NCIE-008, to be | (to be assigned) |
| architecture | assigned)        |                  |

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-1-1: Confirm the NCIE-015 formal title reconciliation (§0.1 — corrected to 'DevSecOps, Infrastructure & Deployment Specification' against Master Controlled Register v2.0), acceptance/sign-off authority, and operations ownership (skeletal Preliminary Human Review Theme 1).

Traceability: NCIE-001–014, especially NCIE-004/009–014.

# DevSecOps & Operational Doctrine

Define common operational principles governing every platform domain this document specifies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/008/009.

## 2.0 Salient Things to Note

Secure automation, reproducibility, traceability, least privilege and recoverability remain Human-governed — automation never acquires authority merely by executing faster.

### 2.1 Secure-by-Default and Reproducibility

Every operational practice is secure-by-default (least privilege, no implicit trust) and reproducible (infrastructure/config changes are versioned and replayable) rather than ad hoc or tribal-knowledge dependent.

### 2.2 Traceability, Least Privilege and Recoverability

Secure automation, reproducibility, traceability, least privilege and recoverability remain Human-governed — automation accelerates execution of an approved procedure; it does not acquire new authority by being automated.

## 2.R RACI / Control Ownership

Table 4. Chapter 2 RACI / control ownership.

| Function                     | Accountable (NCIE-008)                       | Responsible (Operations)                  | Consulted               | Informed                   |
|------------------------------|----------------------------------------------|-------------------------------------------|-------------------------|----------------------------|
| Approve operational doctrine | Accountable Authority Class (to be assigned) | DevSecOps Doctrine Owner (to be assigned) | NCIE-004/008/009 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-2-1: Approve operational doctrine.
- Traceability: NCIE-004/008/009.

# Environment Model & Promotion Boundaries

Define development/test/staging/production separation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/009/014.

## 3.0 Salient Things to Note

Environment promotion ≠ institutional approval; governed data/credentials cannot leak across environments — Table 5 fixes the data/credential policy and prohibited practice for every environment class.

### 3.1 Environment Classes and Isolation

Environment classes (development, test, staging, production, and any specialized class such as a sandboxed Agent-runtime environment) are isolated by network, credential and data boundary; governed data/credentials cannot leak across environments.

### 3.2 Promotion, Refresh and Teardown

Environment promotion ≠ institutional approval — moving a build/config to a higher environment is a technical operation, never itself a release authorization (Chapter 8). Refresh (copying data downward) and teardown follow explicit provenance rules so a lower environment's data state is always traceable to its source and refresh time.

### 3.3 Environment/Promotion Matrix (directive item 57)

The consolidated matrix below maps environment classes to data/credential policy, isolation, deployment source, promotion evidence and prohibited cross-environment practices.

Table 5. Environment/Promotion Matrix.

| Environment Class | Data/Credential Policy                             | Isolation                                           | Deployment Source                 | Promotion Evidence                       | Prohibited Practice                               |
|-------------------|----------------------------------------------------|-----------------------------------------------------|-----------------------------------|------------------------------------------|---------------------------------------------------|
| Development       | Synthetic/masked data only; dev-scoped credentials | Network/credential isolated from staging/production | Feature branch/local build        | N/A — no promotion evidence required     | Production data or production secrets present     |
| Test              | Synthetic/masked data; test-scoped credentials     | Isolated from staging/production                    | CI build artifact                 | CI validation results (Ch.7)             | Production credentials or unmasked regulated data |
| Staging           | Masked/representative data only, per Human Review  | Isolated from production network/credentials        | Release-candidate artifact (Ch.8) | Staging validation/sign-off record       | Direct production credential reuse                |
| Production        | Live governed data under full NCIE-009 controls    | Fully isolated, production credentials only         | Approved release (Ch.8)           | Deployment receipt + health verification | Untested/unapproved artifact deployment           |

## 3.R RACI / Control Ownership

Table 6. Chapter 3 RACI / control ownership.

| Function | Accountable (NCIE-008) | Responsible (Operations) | Consulted | Informed |
|----------|------------------------|--------------------------|-----------|----------|
|----------|------------------------|--------------------------|-----------|----------|

|                                       |                                              |                                    |                         |                            |
|---------------------------------------|----------------------------------------------|------------------------------------|-------------------------|----------------------------|
| Confirm environment topology/data use | Accountable Authority Class (to be assigned) | Environment Owner (to be assigned) | NCIE-004/009/014 owners | Assurance/Audit (NCIE-008) |
|---------------------------------------|----------------------------------------------|------------------------------------|-------------------------|----------------------------|

**Human Review Focus**

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-3-1: Confirm environment topology/data use (Table 5).

*Traceability: NCIE-004/009/014.*

# Infrastructure-as-Code & Configuration Realization

Define reproducible infrastructure/config deployment.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/008/009.

## 4.0 Salient Things to Note

CONFIGURATION DEPLOYED ≠ POLICY APPROVED — every IaC change goes through review before plan/apply, and drift is detected and reconciled through the same reviewed path, never silently.

### 4.1 IaC Modules, State and Review

Infrastructure and configuration are defined as versioned IaC modules with explicit state tracking; every change goes through review before plan/apply, and plan output is inspected before apply rather than applied blind.

### 4.2 Drift, Rollback and Secret References

CONFIGURATION DEPLOYED ≠ POLICY APPROVED — deploying a configuration change operationalizes it; it does not itself constitute institutional policy approval. Drift from the declared state is detected (Chapter 32) and reconciled through the same reviewed path; IaC references secrets by identifier only, never embeds secret values (Chapter 14).

### 4.R RACI / Control Ownership

Table 7. Chapter 4 RACI / control ownership.

| Function                                   | Accountable (NCIE-008)                       | Responsible (Operations)   | Consulted               | Informed                   |
|--------------------------------------------|----------------------------------------------|----------------------------|-------------------------|----------------------------|
| Confirm IaC/config technology and approval | Accountable Authority Class (to be assigned) | IaC Owner (to be assigned) | NCIE-004/008/009 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR15-4-1: Confirm IaC/config technology and approval (no product mandated by this document per directive item 9).

Traceability: NCIE-004/008/009.

# Source Control, Branching & Change Integration

Define source/change integration controls.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/008.

## 5.0 Salient Things to Note

Merge rights ≠ production release or policy authority — source-control permission and release authorization (Chapter 8) remain distinct even for the same individual.

### 5.1 Repositories, Branching and Reviews

Repository structure, branching model and mandatory review are stated as principles; every protected branch requires review before merge, with the review itself recorded as evidence.

### 5.2 Merge Evidence and the Release Boundary

Merge rights ≠ production release or policy authority — the ability to merge code is a source-control permission distinct from the release authorization defined in Chapter 8; merge evidence (who, what, when, review outcome) is retained for traceability.

### 5.R RACI / Control Ownership

Table 8. Chapter 5 RACI / control ownership.

| Function                     | Accountable (NCIE-008)                       | Responsible (Operations)              | Consulted           | Informed                   |
|------------------------------|----------------------------------------------|---------------------------------------|---------------------|----------------------------|
| Confirm source-control model | Accountable Authority Class (to be assigned) | Source Control Owner (to be assigned) | NCIE-004/008 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR15-5-1: Confirm source-control model; does not block the review-before-merge baseline already defined in the interim.

Traceability: NCIE-004/008.

# Build, Artifact & Supply-Chain Pipeline

Define build/artifact production.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/009.

## 6.0 Salient Things to Note

BUILD SUCCESS ≠ RELEASE AUTHORIZATION — a successful, attested, vulnerability-gated build produces an eligible artifact, never an authorized release.

### 6.1 Build, Dependencies and Artifact Identity

BUILD SUCCESS ≠ RELEASE AUTHORIZATION — a successful build produces a versioned, identity-stamped artifact; it does not itself authorize deployment. Dependency resolution is reproducible and recorded.

### 6.2 SBOM, Attestation, Vulnerability Gates and Provenance

Every artifact carries a Software Bill of Materials (SBOM) concept, a build-attestation reference, and passes vulnerability gates before being eligible for CI validation (Chapter 7); provenance links every artifact back to its source commit and build environment.

### 6.R RACI / Control Ownership

Table 9. Chapter 6 RACI / control ownership.

| Function                                    | Accountable (NCIE-008)                       | Responsible (Operations)                  | Consulted           | Informed                   |
|---------------------------------------------|----------------------------------------------|-------------------------------------------|---------------------|----------------------------|
| Confirm build/artifact technology and gates | Accountable Authority Class (to be assigned) | Build/Supply-Chain Owner (to be assigned) | NCIE-004/009 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-6-1: Confirm build/artifact technology and gates.

Traceability: NCIE-004/009.

# CI Validation & Pre-Release Gates

Define automated pre-release validation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/008/009/016.

## 7.0 Salient Things to Note

PIPELINE PASS ≠ PRODUCTION ACCEPTANCE — every gate exception is explicit, recorded and time-bound, never a silent override.

### 7.1 Static, Unit, Integration and Security/Policy Checks

CI validation runs static analysis, unit/integration test interfaces, and security/policy/artifact checks against every candidate build; each check’s pass/fail outcome is recorded as gate evidence.

### 7.2 Pipeline Pass and the Production-Acceptance Boundary

PIPELINE PASS ≠ PRODUCTION ACCEPTANCE — passing every CI gate makes a build eligible for release consideration (Chapter 8); it does not itself constitute production acceptance. Gate exceptions require explicit, recorded, time-bound approval rather than a silent override.

### 7.R RACI / Control Ownership

Table 10. Chapter 7 RACI / control ownership.

| Function                           | Accountable (NCIE-008)                       | Responsible (Operations)             | Consulted                   | Informed                   |
|------------------------------------|----------------------------------------------|--------------------------------------|-----------------------------|----------------------------|
| Confirm mandatory gates/exceptions | Accountable Authority Class (to be assigned) | CI Validation Owner (to be assigned) | NCIE-004/008/009/016 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-7-1: Confirm mandatory gates/exceptions.

Traceability: NCIE-004/008/009/016.

# Release Management & Deployment Authorization

Define release authorization/deployment initiation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008/004.

## 8.0 Salient Things to Note

DEPLOYED ≠ APPROVED; approval binds target/scope/version — Table 11 fixes eight distinct release states so none is ever mistaken for institutional acceptance.

### 8.1 Release Candidate, Approval and Change Record

DEPLOYED ≠ APPROVED — a release candidate becomes deployable only after an explicit approval act that binds a specific target, scope and version; the approval, target, scope and version are recorded together as one change record, never approved in the abstract.

### 8.2 Deployment Receipt and Rollback Trigger

Every deployment produces a receipt (what, where, when, by whom, under which approval) and carries a pre-defined rollback trigger condition, so a failing deployment has an established path back to the last known-good state rather than an improvised one.

### 8.3 Release-Control Matrix (directive item 58)

The consolidated matrix below maps build, CI validation, release candidate, Human/Governance authorization, deployment, health verification, rollback and final acceptance as distinct states, so no state is mistaken for another.

Table 11. Release-Control Matrix.

| State                          | Meaning                                                    | Authority                                  | Never Implies                                                         |
|--------------------------------|------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------|
| Build                          | Artifact produced from source (Ch.6)                       | Automated, no institutional authority      | Release authorization                                                 |
| CI validation passed           | All mandatory gates passed (Ch.7)                          | Automated, no institutional authority      | Production acceptance                                                 |
| Release candidate              | Build eligible for release consideration                   | Automated eligibility only                 | Approval to deploy                                                    |
| Human/Governance authorization | Explicit approval binding target/scope/version             | NCIE-008 accountable authority             | That deployment has occurred                                          |
| Deployed                       | Artifact running in target environment                     | Technical execution of an approval         | That the release is approved or correct                               |
| Health verification            | Post-deployment health checks pass                         | Automated/operator observation             | Correct business outcome (HEALTHY SERVICE ≠ CORRECT BUSINESS OUTCOME) |
| Rollback                       | Reversion to last known-good state                         | Pre-authorized trigger or incident command | Historical erasure of the rolled-back state                           |
| Final acceptance               | Institutional sign-off that the release meets its criteria | NCIE-008 accountable authority             | N/A — this is the terminal state                                      |

8.R RACI / Control Ownership

Table 12. Chapter 8 RACI / control ownership.

| Function                                 | Accountable (NCIE-008)                       | Responsible (Operations)         | Consulted           | Informed                   |
|------------------------------------------|----------------------------------------------|----------------------------------|---------------------|----------------------------|
| Confirm release authority/change classes | Accountable Authority Class (to be assigned) | Release Manager (to be assigned) | NCIE-004/008 owners | Assurance/Audit (NCIE-008) |

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-8-1: Confirm release authority/change classes (Table 11).

Traceability: NCIE-008/004.

# Deployment Strategies & Rollout Control

Define safe rollout patterns.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/011/014.

## 9.0 Salient Things to Note

Partial rollout cannot hide semantic/version mismatch; rollback/reconciliation remain explicit — a mixed-version rollout must stay contract-compatible for its whole duration.

### 9.1 Rolling, Blue-Green, Canary and Feature Flags

Rolling, blue-green and canary rollout concepts, together with feature-flag gating, are stated without mandating a specific platform; sequencing and health gates determine progression between rollout stages.

### 9.2 Compatibility, Rollback and Reconciliation

Partial rollout cannot hide a semantic or version mismatch — a rollout that mixes old and new versions must remain contract-compatible for the duration (NCIE-011); rollback and post-rollout reconciliation remain explicit rather than assumed complete.

### 9.R RACI / Control Ownership

Table 13. Chapter 9 RACI / control ownership.

| Function                     | Accountable (NCIE-008)                       | Responsible (Operations)                | Consulted               | Informed                   |
|------------------------------|----------------------------------------------|-----------------------------------------|-------------------------|----------------------------|
| Confirm permitted strategies | Accountable Authority Class (to be assigned) | Rollout Strategy Owner (to be assigned) | NCIE-004/011/014 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR15-9-1: Confirm permitted strategies; does not block the compatibility/rollback discipline already defined in the interim.
- Traceability: NCIE-004/011/014.

# Runtime Platform & Workload Scheduling

Define application workload infrastructure.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/009.

## 10.0 Salient Things to Note

Scheduler health/placement does not create business authority — a healthy, scheduled workload says nothing about the correctness of what it computes.

## 10.1 Compute Pools, Workload Classes and Scheduling

Compute pools and workload classes are scheduled with explicit resource ceilings and isolation boundaries; scheduler health/placement does not create business authority — a workload being scheduled and healthy says nothing about the correctness of what it computes.

## 10.2 Health, Restart and Lifecycle

Health checks, restart policy and workload lifecycle (start, stop, scale, replace) are stated per workload class; a restarted or rescheduled workload resumes under the same authorization boundary it held before, never an expanded one.

## 10.R RACI / Control Ownership

Table 14. Chapter 10 RACI / control ownership.

| Function                           | Accountable (NCIE-008)                       | Responsible (Operations)                | Consulted           | Informed                   |
|------------------------------------|----------------------------------------------|-----------------------------------------|---------------------|----------------------------|
| Confirm runtime technology/classes | Accountable Authority Class (to be assigned) | Runtime Platform Owner (to be assigned) | NCIE-004/009 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR15-10-1: Confirm runtime technology/classes (no product mandated by this document).

Traceability: NCIE-004/009.

# Agent Runtime & Sandbox Infrastructure

Define physical Agent runtime/sandbox.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-007/009/004.

## 11.0 Salient Things to Note

Agent infrastructure cannot expand Agent authority or security ceilings — sandbox boundaries and resource ceilings apply identically to any generated-code execution path.

### 11.1 Runtime Pools, Sandbox and Ephemeral Lifecycle

Agent runtime pools and sandbox isolation (filesystem/process/network boundaries) are ephemeral by default, with cleanup guaranteed at session/task end; Agent infrastructure cannot expand Agent authority or security ceilings beyond what NCIE-007/009 already grant.

### 11.2 Generated-Code Activation and Resource Ceilings

Any generated-code execution path is scoped by the same sandbox boundary and resource ceilings as the rest of the Agent runtime; activation of generated code never bypasses the NCIE-007 approval/validation chain governing what an Agent may execute.

### 11.R RACI / Control Ownership

Table 15. Chapter 11 RACI / control ownership.

| Function                                              | Accountable (NCIE-008)                       | Responsible (Operations)             | Consulted               | Informed                   |
|-------------------------------------------------------|----------------------------------------------|--------------------------------------|-------------------------|----------------------------|
| Confirm sandbox/runtime and generated-code activation | Accountable Authority Class (to be assigned) | Agent Runtime Owner (to be assigned) | NCIE-004/007/009 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-11-1: Confirm sandbox/runtime and generated-code activation.

Traceability: NCIE-007/009/004.

NCIE-015 — Chapter 12 Expansion

# Network, Ingress, Egress & Service Connectivity Operations

Define network realization.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009/004.

## 12.0 Salient Things to Note

Network reachability ≠ application authorization — being reachable never substitutes for the application-layer authorization check a request still requires.

## 12.1 Ingress, Egress and Service Discovery

Ingress, egress, service connectivity and discovery are operationally realized per NCIE-009 policy; network reachability ≠ application authorization — being network-reachable never substitutes for the application-layer authorization check a request still requires.

## 12.2 Segmentation, Policy Realization and Network Change

Segmentation enforces the boundaries NCIE-009 defines; network policy changes follow the same change-management discipline as any other production change (Chapter 25), not an informal network-team exception path.

## 12.R RACI / Control Ownership

Table 16. Chapter 12 RACI / control ownership.

| Function                        | Accountable (NCIE-008)                       | Responsible (Operations)                  | Consulted           | Informed                   |
|---------------------------------|----------------------------------------------|-------------------------------------------|---------------------|----------------------------|
| Confirm topology/egress classes | Accountable Authority Class (to be assigned) | Network Operations Owner (to be assigned) | NCIE-004/009 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-12-1: Confirm topology/egress classes.

Traceability: NCIE-009/004.

# IAM, Workload Identity & Privileged Operations

Operationalize identity/privilege controls.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-009/008.

## 13.0 Salient Things to Note

OPERATOR PRIVILEGE ≠ INSTITUTIONAL DECISION AUTHORITY; infrastructure access ≠ content authority — every privileged session is attributable regardless of operator seniority.

### 13.1 Admin/Workload Identity, Elevation and Break-Glass

Admin and workload identities, just-in-time (JIT) elevation concept and break-glass procedures follow NCIE-009 policy; OPERATOR PRIVILEGE ≠ INSTITUTIONAL DECISION AUTHORITY — an operator's ability to act on infrastructure never confers institutional decision-making authority over content.

### 13.2 Session Controls, Revocation and Privileged Audit

INFRASTRUCTURE ACCESS ≠ CONTENT AUTHORITY — session controls, timely revocation and privileged-action audit apply uniformly regardless of an operator's seniority; every privileged session is attributable.

### 13.R RACI / Control Ownership

Table 17. Chapter 13 RACI / control ownership.

| Function                                  | Accountable (NCIE-008)                       | Responsible (Operations)              | Consulted           | Informed                   |
|-------------------------------------------|----------------------------------------------|---------------------------------------|---------------------|----------------------------|
| Confirm privilege/on-call emergency model | Accountable Authority Class (to be assigned) | IAM Operations Owner (to be assigned) | NCIE-008/009 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-13-1: Confirm privilege/on-call emergency model.
- Traceability: NCIE-009/008.

NCIE-015 — Chapter 14 Expansion

# Secrets, Certificates & Key Operations

Define secret/certificate operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-009/004.

## 14.0 Salient Things to Note

Workload secret availability ≠ human disclosure authority — a workload holding a secret to function never entitles a human operator to view it.

### 14.1 Secret Delivery, Rotation and Certificates

Secrets and certificates are delivered to workloads by reference, not embedded in code or IaC; rotation cadence follows NCIE-009 policy, and workload secret availability ≠ human disclosure authority — a workload holding a secret to function does not mean a human operator is entitled to view it.

### 14.2 Key References, Revocation and Emergency Rotation

Key references bind to their workload scope; revocation and emergency rotation procedures are defined so a compromised secret/certificate/key can be rotated without an extended production outage, and every rotation event is logged.

### 14.R RACI / Control Ownership

Table 18. Chapter 14 RACI / control ownership.

| Function                        | Accountable (NCIE-008)                       | Responsible (Operations)                      | Consulted           | Informed                   |
|---------------------------------|----------------------------------------------|-----------------------------------------------|---------------------|----------------------------|
| Confirm custody/rotation policy | Accountable Authority Class (to be assigned) | Secrets/Key Operations Owner (to be assigned) | NCIE-004/009 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-14-1: Confirm custody/rotation policy.
- Traceability: NCIE-009/004.

# Database & Storage Deployment Operations

Operationalize NCIE-014 persistence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-014.

## 15.0 Salient Things to Note

Storage operations cannot redefine semantics, retention, residency or authoritative reads — every operational action stays strictly within the NCIE-014 persistence architecture.

### 15.1 Provisioning, Migrations and Backup Jobs

Storage operations cannot redefine semantics, retention, residency or authoritative reads — provisioning, migration execution (per NCIE-014 Ch.25-26) and backup job scheduling operate strictly within the persistence architecture NCIE-014 already defines.

### 15.2 Replicas, Indexes, Capacity and Maintenance

Replica/index operational maintenance and capacity management follow NCIE-014's replication (Ch.18) and residency (Ch.20) requirements; maintenance windows and restore operations are coordinated with Chapters 25 and 27 respectively rather than executed ad hoc.

### 15.R RACI / Control Ownership

Table 19. Chapter 15 RACI / control ownership.

| Function                              | Accountable<br>(NCIE-008)                    | Responsible<br>(Operations)                | Consulted      | Informed                   |
|---------------------------------------|----------------------------------------------|--------------------------------------------|----------------|----------------------------|
| Confirm ownership/maintenance windows | Accountable Authority Class (to be assigned) | Database Operations Owner (to be assigned) | NCIE-014 owner | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-15-1: Confirm ownership/maintenance windows.
- Traceability: NCIE-014.

# API Gateway, Event Bus & Connector Operations

Operationalize NCIE-011 integration infrastructure.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-011.

## 16.0 Salient Things to Note

Healthy integration infrastructure ≠ correct business outcome — a responsive gateway says nothing about whether the underlying business data is correct.

## 16.1 Gateway, Events and Connectors

API gateway, event bus and connector infrastructure are operated per NCIE-011 contracts; healthy integration infrastructure ≠ correct business outcome — a gateway returning 200 OK says nothing about whether the underlying business data is correct.

## 16.2 Schema Deployment, Lag, Retries and Quarantine

Schema deployment for events/connectors follows NCIE-011 versioning; consumer lag, retry policy and quarantine (for poison messages) are operated with reconciliation back to source so no message is silently dropped without record.

## 16.R RACI / Control Ownership

Table 20. Chapter 16 RACI / control ownership.

| Function                              | Accountable (NCIE-008)                       | Responsible (Operations)                      | Consulted      | Informed                   |
|---------------------------------------|----------------------------------------------|-----------------------------------------------|----------------|----------------------------|
| Confirm platform ownership/thresholds | Accountable Authority Class (to be assigned) | Integration Operations Owner (to be assigned) | NCIE-011 owner | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR15-16-1: Confirm platform ownership/thresholds.

Traceability: NCIE-011.

# ARGUS, Model Gateway, Tool Gateway & VPF Operations

Define operations for core intelligence services.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005/007/009/011.

## 17.0 Salient Things to Note

Availability cannot alter ARGUS/Model/Tool/VPF authority or validation semantics — a highly available Tool Gateway does not relax the NCIE-007 approval a Tool call still requires.

## 17.1 Deployment, Scaling and Routing

ARGUS, Model Gateway, Tool Gateway and VPF services are deployed, scaled and routed operationally; availability cannot alter ARGUS/Model/Tool/VPF authority or validation semantics — a highly available Tool Gateway does not relax the NCIE-007 approval a Tool call still requires.

## 17.2 Provider/Tool Connectivity, VPF Availability and Rollback

Provider/Tool connectivity health and VPF availability are monitored operationally; rollback of a core intelligence service follows the same release-control discipline (Chapter 8) as any other production service.

## 17.R RACI / Control Ownership

Table 21. Chapter 17 RACI / control ownership.

| Function                          | Accountable (NCIE-008)                       | Responsible (Operations)                        | Consulted                   | Informed                   |
|-----------------------------------|----------------------------------------------|-------------------------------------------------|-----------------------------|----------------------------|
| Confirm service/on-call ownership | Accountable Authority Class (to be assigned) | Core Services Operations Owner (to be assigned) | NCIE-005/007/009/011 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-17-1: Confirm service/on-call ownership.

Traceability: NCIE-005/007/009/011.

# Functional Module Operations

Define operational realization of NCIE-013.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-013.

## 18.0 Salient Things to Note

Operations cannot alter domain Findings, rules or thresholds outside governed change — module configuration rollout follows the same change-management path as any other production change.

## 18.1 Module Deployment, Schedules and Dependencies

NCIE-013 domain modules are deployed, scheduled and dependency-managed operationally; operations cannot alter domain Findings, rules or thresholds outside a governed NCIE-013/NCIE-008 change.

## 18.2 Configuration Rollout, Freshness and Health

Module configuration rollout (rule/KPI updates) follows the same change-management path as any other production change; freshness and health of each module's data pipeline are monitored so a stale module is detected rather than silently trusted.

## 18.R RACI / Control Ownership

Table 22. Chapter 18 RACI / control ownership.

| Function                         | Accountable (NCIE-008)                       | Responsible (Operations)                 | Consulted      | Informed                   |
|----------------------------------|----------------------------------------------|------------------------------------------|----------------|----------------------------|
| Confirm module support ownership | Accountable Authority Class (to be assigned) | Module Operations Owner (to be assigned) | NCIE-013 owner | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-18-1: Confirm module support ownership.

Traceability: NCIE-013.

# UX / Frontend Delivery Operations

Define frontend/design-system delivery operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-012/009/011.

## 19.0 Salient Things to Note

Frontend deployment cannot weaken authorization or semantic state presentation — a faster page must still present the same authorization and state semantics as the reference build.

### 19.1 Frontend Build, Assets and Edge/CDN Delivery

Frontend build, asset bundling and edge/CDN delivery are operated to serve NCIE-012's design system; frontend deployment cannot weaken authorization or semantic state presentation — a faster-loading page must still present the same NCIE-009 authorization and NCIE-013 state semantics as the reference build.

### 19.2 Config, Feature Delivery, Cache Invalidation and Rollback

Frontend configuration and feature-flag delivery follow the same change discipline as backend releases; cache invalidation on deploy prevents stale UI from presenting outdated authorization or state, and rollback restores the prior frontend version cleanly.

### 19.R RACI / Control Ownership

Table 23. Chapter 19 RACI / control ownership.

| Function                  | Accountable (NCIE-008)                       | Responsible (Operations)                   | Consulted               | Informed                   |
|---------------------------|----------------------------------------------|--------------------------------------------|-------------------------|----------------------------|
| Confirm frontend topology | Accountable Authority Class (to be assigned) | Frontend Operations Owner (to be assigned) | NCIE-009/011/012 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR15-19-1: Confirm frontend topology; does not block the cache-invalidation/rollback discipline already defined in the interim.

Traceability: NCIE-012/009/011.

# Observability Architecture & Telemetry Operations

Define metrics/logs/traces/health operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/009/010.

## 20.0 Salient Things to Note

TELEMETRY ≠ AUDIT ≠ PROVENANCE; healthy telemetry ≠ business correctness — Table 24 draws an explicit line between every telemetry type and the one authoritative audit/provenance class.

### 20.1 Metrics, Logs, Traces and Correlation

Metrics, logs and traces are correlated by request/trace identity; dashboards surface health and SLO indicators (Chapter 21) for operational use.

### 20.2 Access, Retention and the Audit/Provenance Boundary

TELEMETRY ≠ AUDIT ≠ PROVENANCE — telemetry access follows NCIE-009 authorization and retention follows an explicit interface; healthy telemetry ≠ business correctness, and telemetry is never substituted for the audit/provenance records Chapter 31 and NCIE-010 require.

### 20.3 Observability Matrix (directive item 62)

The consolidated matrix below maps telemetry types to operational purpose, access, retention interface and explicit separation from audit/provenance/business truth.

Table 24. Observability Matrix.

| Telemetry Type                             | Operational Purpose                            | Access                        | Retention Interface                                       | Separation Statement                                                 |
|--------------------------------------------|------------------------------------------------|-------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------|
| Metrics                                    | Health/SLO indicator tracking (Ch.21)          | NCIE-009 authorized operators | Tiered per Ch.21 SLO needs                                | Not audit/provenance; not business correctness                       |
| Logs                                       | Debugging, operational troubleshooting         | NCIE-009 authorized operators | Per retention interface, distinct from NCIE-010 retention | Not audit/provenance; may contain operational detail only            |
| Traces                                     | Request/latency correlation across services    | NCIE-009 authorized operators | Short-to-medium tiered retention                          | Not audit/provenance; not a Finding basis                            |
| Alerts (Ch.22)                             | Operational notification of a threshold breach | On-call/operations            | Per incident-record retention (Ch.31)                     | ALERT ≠ INCIDENT; not a Finding                                      |
| Audit/provenance records (NCIE-010, Ch.31) | Historical accountability, Evidence basis      | NCIE-010 governed access      | NCIE-010 retention policy                                 | Authoritative — the only class this table treats as audit/provenance |

20.R RACI / Control Ownership

Table 25. Chapter 20 RACI / control ownership.

| Function                                 | Accountable (NCIE-008)                       | Responsible (Operations)             | Consulted               | Informed                   |
|------------------------------------------|----------------------------------------------|--------------------------------------|-------------------------|----------------------------|
| Confirm observability platform/retention | Accountable Authority Class (to be assigned) | Observability Owner (to be assigned) | NCIE-004/009/010 owners | Assurance/Audit (NCIE-008) |

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR15-20-1: Confirm observability platform/retention (Table 24).

Traceability: NCIE-004/009/010.

# Service-Level Objectives, Indicators & Error Budgets

Define reliability measurement.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-002/004/013/014.

## 21.0 Salient Things to Note

SLO attainment ≠ business correctness/acceptance; numeric targets require authority — no specific SLO/SLI value is fixed by this document.

### 21.1 SLIs, SLOs and Error Budgets

SLI/SLO/error-budget framework is defined conceptually (availability, latency, freshness, dependency health); SLO attainment ≠ business correctness/acceptance — meeting an SLO says nothing about the correctness of the business outcome it measures adjacent to.

### 21.2 Numeric Targets Require Authority

No specific SLO/SLI numeric target is fixed by this document; every numeric target requires Human Review confirmation, consistent with directive item 10's no-numeric-invention rule.

### 21.R RACI / Control Ownership

Table 26. Chapter 21 RACI / control ownership.

| Function                      | Accountable (NCIE-008)                       | Responsible (Operations)   | Consulted                   | Informed                   |
|-------------------------------|----------------------------------------------|----------------------------|-----------------------------|----------------------------|
| Confirm SLO catalogue/targets | Accountable Authority Class (to be assigned) | SLO Owner (to be assigned) | NCIE-002/004/013/014 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-21-1: Confirm SLO catalogue/targets (no numeric value invented by this document).

Traceability: NCIE-002/004/013/014.

# Alerting, On-Call & Operational Escalation

Define alerting/on-call response.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/008/012.

## 22.0 Salient Things to Note

ALERT ≠ INCIDENT; alert severity ≠ root cause/Finding — an alert becomes an incident only through the Chapter 23 declaration act.

### 22.1 Alert Rules, Routing and On-Call

ALERT ≠ INCIDENT — an alert is a threshold-driven notification; it becomes an incident only through the declaration act defined in Chapter 23. Alert routing, on-call rotation and acknowledgement are operated per approved escalation policy.

### 22.2 Suppression, Maintenance and Handoff

Alert suppression during approved maintenance windows (Chapter 25) is time-bound and reversible; on-call handoff carries forward open items so nothing is silently dropped between shifts.

### 22.R RACI / Control Ownership

Table 27. Chapter 22 RACI / control ownership.

| Function                   | Accountable (NCIE-008)                       | Responsible (Operations)        | Consulted               | Informed                   |
|----------------------------|----------------------------------------------|---------------------------------|-------------------------|----------------------------|
| Confirm on-call/thresholds | Accountable Authority Class (to be assigned) | Alerting Owner (to be assigned) | NCIE-004/008/012 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-22-1: Confirm on-call/thresholds.

Traceability: NCIE-004/008/012.

# Incident Management & Command

Define incident lifecycle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009/008/010.

## 23.0 Salient Things to Note

INCIDENT DECLARED ≠ ROOT CAUSE ESTABLISHED; incident command ≠ policy-exception authority — Table 28 fixes reconciliation and Evidence requirements for every incident/recovery state.

### 23.1 Detection, Declaration and Triage

INCIDENT DECLARED ≠ ROOT CAUSE ESTABLISHED — declaration opens the incident lifecycle (detection, triage, role assignment) without asserting a root cause; triage assigns severity and incident-command roles per approved policy.

### 23.2 Containment, Communications and Closure

Incident command ≠ policy-exception authority — the incident commander directs response within pre-approved runbook/emergency-change boundaries (Chapters 24-25), never granting itself new policy authority. Timeline, containment actions and closure are all preserved as Evidence (Chapter 31).

### 23.3 Incident/Recovery Matrix (directive item 61)

The consolidated matrix below maps incident/failover/restore/DR states to command authority, containment, current-state reconciliation, Evidence preservation and acceptance.

Table 28. Incident/Recovery Matrix.

| State                     | Command Authority                                  | Containment Action                  | Current-State Reconciliation                                         | Evidence Preservation      | Acceptance                                       |
|---------------------------|----------------------------------------------------|-------------------------------------|----------------------------------------------------------------------|----------------------------|--------------------------------------------------|
| Incident declared         | Incident Commander (per approved on-call policy)   | Per runbook (Ch.24)                 | N/A at declaration                                                   | Timeline opened            | N/A — lifecycle in progress                      |
| Contained                 | Incident Commander                                 | Active containment per runbook      | Reconcile affected systems' current state before further action      | Containment actions logged | N/A — recovery pending                           |
| Failover executed (Ch.28) | DR/Incident Commander per NCIE-014 Ch.24 authority | Fencing/split-brain prevention      | Reconcile identity/authorization/Evidence state at promotion         | Failover event logged      | Post-failover health/reconciliation confirmation |
| Restore executed (Ch.27)  | Incident Commander or Backup/Restore Owner         | Isolation during restore validation | Reconcile current identity/eligibility/policy before serving traffic | Restore event logged       | Restore-validation record                        |
| Recovery complete         | Incident Commander confirms                        | N/A                                 | Final reconciliation confirmed                                       | Full timeline closed       | Closure sign-off                                 |
| Post-incident review      | Accountable Authority (NCIE-008)                   | N/A                                 | N/A                                                                  | Lessons-learned record     | Institutional acceptance of findings/remediation |

23.R RACI / Control Ownership

Table 29. Chapter 23 RACI / control ownership.

| Function                            | Accountable (NCIE-008)                       | Responsible (Operations)                 | Consulted               | Informed                   |
|-------------------------------------|----------------------------------------------|------------------------------------------|-------------------------|----------------------------|
| Confirm incident authority/severity | Accountable Authority Class (to be assigned) | Incident Commander role (to be assigned) | NCIE-008/009/010 owners | Assurance/Audit (NCIE-008) |

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-23-1: Confirm incident authority/severity (Table 28).

Traceability: NCIE-009/008/010.

# Runbooks, Playbooks & Operational Procedures

Define governed runbook architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009–014 handovers.

## 24.0 Salient Things to Note

RUNBOOK EXECUTION ≠ POLICY EXCEPTION AUTHORITY — Table 30 fixes an explicit stop/escalation condition for every runbook so automation never expands beyond its approved scope.

### 24.1 Runbook Identity, Trigger and Prerequisites

RUNBOOK EXECUTION ≠ POLICY EXCEPTION AUTHORITY — a runbook executes a pre-approved procedure; it does not grant its executor authority beyond what the runbook itself was approved to do. Every runbook has an identity, trigger condition and stated prerequisites.

### 24.2 Steps, Verification, Rollback and Evidence

Runbook steps, verification checkpoints, rollback path, escalation condition, evidence capture, version and owner are all explicit; an undocumented ad hoc procedure is never substituted for an approved runbook during a live incident.

### 24.3 Runbook Matrix (directive item 60)

The consolidated matrix below maps operational scenarios to runbook trigger, prerequisites, authority boundary, stop/escalation conditions, evidence and rollback.

Table 30. Runbook Matrix.

| Operational Scenario                | Trigger                                    | Prerequisites                      | Authority Boundary             | Stop/Escalation Condition                                              | Evidence/Rollback                                                            |
|-------------------------------------|--------------------------------------------|------------------------------------|--------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Standard deployment (Ch.8-9)        | Approved release                           | CI gates passed, approval recorded | Release Manager                | Health-gate failure → automatic rollback trigger                       | Deployment receipt; rollback to last known-good                              |
| Secret/certificate rotation (Ch.14) | Scheduled or emergency                     | Key-reference validated            | Secrets/Key Operations Owner   | Binding failure → escalate to IAM Operations Owner                     | Rotation event logged; revert to prior key reference if bound workload fails |
| Failover (Ch.28)                    | DR declaration                             | Fencing pre-checks passed          | Incident/DR Commander          | Split-brain risk detected → halt and escalate to Accountable Authority | Failover event logged; fallback runbook available                            |
| Restore (Ch.27)                     | Data-loss/corruption incident              | Backup catalog validated           | Backup/Restore Owner           | Integrity-check failure → escalate, do not serve restored data         | Restore-validation record                                                    |
| Emergency change (Ch.25)            | Active incident requiring immediate change | Incident Commander authorization   | Incident Commander, time-boxed | Change exceeds pre-approved scope → escalate to                        | Emergency-change record; mandatory post-review                               |

Accountable  
Authority

24.R RACI / Control Ownership

Table 31. Chapter 24 RACI / control ownership.

| Function                             | Accountable (NCIE-008)                       | Responsible (Operations)       | Consulted                   | Informed                   |
|--------------------------------------|----------------------------------------------|--------------------------------|-----------------------------|----------------------------|
| Confirm runbook ownership/automation | Accountable Authority Class (to be assigned) | Runbook Owner (to be assigned) | NCIE-009-014 chapter owners | Assurance/Audit (NCIE-008) |

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-24-1: Confirm runbook ownership/automation (Table 30).

Traceability: NCIE-009–014 handovers.

# Change, Maintenance & Emergency Operations

Define routine/standard/emergency change.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008/009.

## 25.0 Salient Things to Note

EMERGENCY CHANGE ≠ PERMANENT POLICY CHANGE — every emergency change requires mandatory post-review and either ratification or reversion, never silent adoption.

### 25.1 Change Classes, Windows and Approval

Routine, standard and emergency change classes each carry their own approval requirement and maintenance-window policy; freeze periods (e.g. around critical events) suspend routine change without suspending emergency-change capability.

### 25.2 Emergency Change and Post-Review

EMERGENCY CHANGE ≠ PERMANENT POLICY CHANGE — an emergency change is temporary and governed, requiring mandatory post-incident review and either formal ratification into standard policy or reversion; it never silently becomes the new baseline.

### 25.R RACI / Control Ownership

Table 32. Chapter 25 RACI / control ownership.

| Function                                          | Accountable<br>(NCIE-008)                          | Responsible<br>(Operations)                       | Consulted              | Informed                      |
|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------|------------------------|-------------------------------|
| Confirm<br>classes/windows/emergency<br>authority | Accountable<br>Authority Class (to<br>be assigned) | Change<br>Management<br>Owner (to be<br>assigned) | NCIE-008/009<br>owners | Assurance/Audit<br>(NCIE-008) |

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-25-1: Confirm classes/windows/emergency authority.

Traceability: NCIE-008/009.

# Capacity, Scaling & Performance Operations

Define capacity/scaling operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-002/004/014.

## 26.0 Salient Things to Note

AUTOSCALING ≠ CAPACITY ASSURANCE; scaling cannot weaken isolation/sovereignty/state correctness — a scale-out event never bypasses the residency controls that would otherwise apply.

## 26.1 Forecasts, Scaling Policies and Quotas

AUTOSCALING ≠ CAPACITY ASSURANCE — autoscaling reacts to observed load; it does not itself guarantee sufficient capacity exists to scale into. Capacity forecasting, scaling-policy concepts and quotas are stated as operational levers, not guarantees.

## 26.2 Saturation, Baselines and Sovereignty Boundary

Load signals and saturation baselines inform scaling decisions; scaling cannot weaken isolation, sovereignty (NCIE-014 Ch.20) or state correctness — a scale-out event never moves data across an approved residency boundary without the same Chapter 12/NCIE-014 controls applying.

## 26.R RACI / Control Ownership

Table 33. Chapter 26 RACI / control ownership.

| Function               | Accountable (NCIE-008)                       | Responsible (Operations)        | Consulted               | Informed                   |
|------------------------|----------------------------------------------|---------------------------------|-------------------------|----------------------------|
| Confirm targets/limits | Accountable Authority Class (to be assigned) | Capacity Owner (to be assigned) | NCIE-002/004/014 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR15-26-1: Confirm targets/limits (no numeric value invented by this document).

Traceability: NCIE-002/004/014.

# Backup, Restore & Data-Protection Operations

Define backup/restore execution.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-014/009/010.

## 27.0 Salient Things to Note

RESTORE ≠ REAUTHORIZATION; restored systems reconcile current policy/identity/eligibility — every restore attempt, successful or failed, is recorded rather than silently retried.

## 27.1 Schedules, Monitoring and Restore

Backup job scheduling and monitoring operationalize NCIE-014 Chapter 23's backup architecture; restore execution follows the recovery sequence NCIE-014 already defines, with isolation during restore validation.

## 27.2 Reconciliation, Evidence and Failure Handling

RESTORE ≠ REAUTHORIZATION — restored systems reconcile current policy, identity and eligibility before being treated as authoritative (NCIE-014 §23.2); restore evidence and failure handling (a failed restore attempt) are both recorded, never silently retried without record.

## 27.R RACI / Control Ownership

Table 34. Chapter 27 RACI / control ownership.

| Function                              | Accountable<br>(NCIE-008)                          | Responsible<br>(Operations)                 | Consulted                  | Informed                      |
|---------------------------------------|----------------------------------------------------|---------------------------------------------|----------------------------|-------------------------------|
| Confirm<br>schedule/RPO/RTO/authority | Accountable<br>Authority Class (to<br>be assigned) | Backup/Restore<br>Owner (to be<br>assigned) | NCIE-009/010/014<br>owners | Assurance/Audit<br>(NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-27-1: Confirm schedule/RPO/RTO/authority (no numeric value invented by this document).

Traceability: NCIE-014/009/010.

# Disaster Recovery, Failover & Business Continuity Operations

Define DR/failover/failback.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-004/014/009.

## 28.0 Salient Things to Note

FAILOVER ≠ REAUTHORIZATION; DR EXERCISE ≠ PRODUCTION RECOVERY PROOF — an exercise demonstrates the exercised path only, never real production-failure recovery.

### 28.1 DR Declaration, Failover and Fencing

FAILOVER ≠ REAUTHORIZATION — DR declaration and failover execution follow NCIE-014 Chapter 24's cross-site persistence architecture; fencing prevents split-brain, and failover priority order is stated explicitly rather than improvised during an incident.

### 28.2 Reconciliation, Failback, Exercises and Lessons

DR EXERCISE ≠ PRODUCTION RECOVERY PROOF — a successful DR exercise demonstrates the exercised path works under exercise conditions; it does not prove production recovery under real failure conditions. Failback reconciliation and lessons-learned are captured after every real or exercised event.

### 28.R RACI / Control Ownership

Table 35. Chapter 28 RACI / control ownership.

| Function                                 | Accountable<br>(NCIE-008)                          | Responsible<br>(Operations)                | Consulted                  | Informed                      |
|------------------------------------------|----------------------------------------------------|--------------------------------------------|----------------------------|-------------------------------|
| Confirm DR<br>authority/topology/cadence | Accountable<br>Authority Class (to<br>be assigned) | DR Operations<br>Owner (to be<br>assigned) | NCIE-004/009/014<br>owners | Assurance/Audit<br>(NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-28-1: Confirm DR authority/topology/cadence.

Traceability: NCIE-004/014/009.

# Vulnerability, Patch & Remediation Operations

Define vulnerability/patch operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009/008/004.

## 29.0 Salient Things to Note

Scanner severity ≠ institutional risk acceptance; exceptions remain governed/expiring — a severity rating informs triage, it never itself accepts the risk.

## 29.1 Discovery, Triage and Patch

Vulnerability discovery, triage and patching follow a defined workflow; scanner severity ≠ institutional risk acceptance — a scanner's severity rating informs triage but never itself constitutes the institutional risk-acceptance decision.

## 29.2 Verification, Exception and Compensating Controls

Remediation verification confirms the vulnerability is actually closed, not merely patched-looking; exceptions remain governed and expiring, with compensating controls stated explicitly rather than left implicit.

## 29.R RACI / Control Ownership

Table 36. Chapter 29 RACI / control ownership.

| Function                                   | Accountable (NCIE-008)                       | Responsible (Operations)                        | Consulted               | Informed                   |
|--------------------------------------------|----------------------------------------------|-------------------------------------------------|-------------------------|----------------------------|
| Confirm remediation targets/risk interface | Accountable Authority Class (to be assigned) | Vulnerability Management Owner (to be assigned) | NCIE-004/008/009 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-29-1: Confirm remediation targets/risk interface (no deadline value invented by this document).

Traceability: NCIE-009/008/004.

# Operational Security Monitoring & Response Integration

Define security-operations interfaces.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009/010.

## 30.0 Salient Things to Note

Security alert/containment ≠ institutional Finding — a triaged and contained security event is never itself a governed Finding until NCIE-008's Finding process establishes one.

### 30.1 Security Alerts, Anomalies and Egress/Exfiltration

Security-alert, anomaly and egress/exfiltration signal triage are operated per NCIE-009 policy; security alert/containment ≠ institutional Finding — a triaged and contained security event is not itself a governed Finding until NCIE-008's Finding process establishes one.

### 30.2 Agent/Tool-Abuse Triage and Evidence Handoff

Agent/Tool-abuse indicators are triaged with the same discipline as any security signal; containment handoff and evidence preservation follow the Chapter 31/NCIE-010 chain so a security response never destroys the record a later Finding process would need.

### 30.R RACI / Control Ownership

Table 37. Chapter 30 RACI / control ownership.

| Function                            | Accountable (NCIE-008)                       | Responsible (Operations)                   | Consulted           | Informed                   |
|-------------------------------------|----------------------------------------------|--------------------------------------------|---------------------|----------------------------|
| Confirm SecOps ownership/escalation | Accountable Authority Class (to be assigned) | Security Operations Owner (to be assigned) | NCIE-009/010 owners | Assurance/Audit (NCIE-008) |

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR15-30-1: Confirm SecOps ownership/escalation.

Traceability: NCIE-009/010.

# Operational Audit, Evidence & Change Provenance

Define accountable operational evidence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-010/008/011.

## 31.0 Salient Things to Note

Telemetry is not audit/provenance; releases/incidents/changes remain historically attributable — every operational record ties an action to a specific approval, actor and time.

## 31.1 Change/Deployment Records and Incident Timeline

Every deployment, change, incident and runbook execution produces a durable record — telemetry is not audit/provenance (Chapter 20); these operational records are the actual audit/provenance basis for operations, retained per NCIE-010 requirements.

## 31.2 Approvals, Evidence and Historical Attributability

Approvals (Chapter 8/25) are recorded alongside the action they authorized, so releases, incidents and changes remain historically attributable to a specific approval, actor and time — never reconstructed after the fact from inference.

## 31.R RACI / Control Ownership

Table 38. Chapter 31 RACI / control ownership.

| Function                        | Accountable (NCIE-008)                       | Responsible (Operations)                 | Consulted               | Informed                   |
|---------------------------------|----------------------------------------------|------------------------------------------|-------------------------|----------------------------|
| Confirm record retention/access | Accountable Authority Class (to be assigned) | Operational Audit Owner (to be assigned) | NCIE-008/010/011 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR15-31-1: Confirm record retention/access (no retention period invented by this document).

Traceability: NCIE-010/008/011.

# Operational Reconciliation, Drift & Configuration Integrity

Define drift/reconciliation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/008/014.

## 32.0 Salient Things to Note

Auto-remediation cannot silently override approved emergency state, exception or current authorization — every repair is proposed, approved, provenance-recorded and post-verified.

## 32.1 Desired/Actual State and Drift Detection

Drift between declared (IaC/Chapter 4) and actual infrastructure state is detected on a scheduled basis; detection alone never triggers an automated write — repair follows the approval path below.

## 32.2 Reconciliation, Repair Approval and Provenance

Auto-remediation cannot silently override an approved emergency state, exception or current authorization — every repair is proposed, approved, provenance-recorded and independently verified after application, mirroring NCIE-014 Chapter 27's discipline for data-layer repair.

## 32.R RACI / Control Ownership

Table 39. Chapter 32 RACI / control ownership.

| Function                            | Accountable (NCIE-008)                       | Responsible (Operations)              | Consulted               | Informed                   |
|-------------------------------------|----------------------------------------------|---------------------------------------|-------------------------|----------------------------|
| Confirm auto-remediation boundaries | Accountable Authority Class (to be assigned) | Reconciliation Owner (to be assigned) | NCIE-004/008/014 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-32-1: Confirm auto-remediation boundaries.

Traceability: NCIE-004/008/014.

# Operational Readiness, Handover & Go-Live Governance

Define readiness evidence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.  
Upstream: NCIE-008/016.

## 33.0 Salient Things to Note

Deployment readiness ≠ Human/Institutional acceptance; blockers/failed tests remain visible — a completed checklist is a precondition for go-live, never the acceptance decision itself.

## 33.1 Readiness Checklist and Ownership

Deployment readiness ≠ Human/Institutional acceptance — a completed technical readiness checklist (runbooks in place, monitoring configured, backup/DR verified, security reviewed, capacity confirmed) is a precondition for go-live, never itself the acceptance decision.

## 33.2 Risks, Blockers and Approvals

Known risks and any failed or skipped test remain visible in the readiness record rather than hidden by an aggregated 'ready' status; go-live approval is a distinct institutional act (NCIE-008) taken with that risk record in view.

## 33.R RACI / Control Ownership

Table 40. Chapter 33 RACI / control ownership.

| Function                           | Accountable (NCIE-008)                       | Responsible (Operations)                 | Consulted           | Informed                   |
|------------------------------------|----------------------------------------------|------------------------------------------|---------------------|----------------------------|
| Confirm go-live authority/evidence | Accountable Authority Class (to be assigned) | Go-Live Readiness Owner (to be assigned) | NCIE-008/016 owners | Assurance/Audit (NCIE-008) |

## Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-33-1: Confirm go-live authority/evidence.

Traceability: NCIE-008/016.

# DevSecOps / Operations Testing & Game-Day Validation

Define operational test/exercise classes.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-016/009/014.

## 34.0 Salient Things to Note

TEST SPECIFIED ≠ TEST EXECUTED; game-day success ≠ proof of every production failure mode — all ten test classes above are SPECIFIED only.

## 34.1 Test Class Catalogue

Ten operational test classes are specified: pipeline, deploy/rollback, secret-rotation, failover, restore, incident-response, capacity, containment, drift and game-day. Each states its objective and acceptance basis; TEST SPECIFIED ≠ TEST EXECUTED.

Table 41. DevSecOps/Operations Test-Class Catalogue.

| Test Class        | Objective                                                                      | Acceptance Basis                          | Status    |
|-------------------|--------------------------------------------------------------------------------|-------------------------------------------|-----------|
| Pipeline          | Verify CI gates correctly block a non-compliant build (Ch.7)                   | Negative-test gate-failure suite          | SPECIFIED |
| Deploy/rollback   | Verify deployment and rollback trigger behave per Table 11 (Ch.8)              | Deploy-then-rollback drill                | SPECIFIED |
| Secret rotation   | Verify rotation completes without workload disruption (Ch.14)                  | Rotation drill with workload health check | SPECIFIED |
| Failover          | Verify cross-site failover/fencing behavior (Ch.28)                            | DR-drill scenario results                 | SPECIFIED |
| Restore           | Verify restore reconciles current state before serving (Ch.27)                 | Restore-validation checklist              | SPECIFIED |
| Incident response | Verify declaration/triage/command roles execute per Table 28 (Ch.23)           | Tabletop or live incident simulation      | SPECIFIED |
| Capacity          | Verify scaling policy behaves under load without breaching sovereignty (Ch.26) | Load/scale test suite                     | SPECIFIED |
| Containment       | Verify security containment handoff preserves Evidence (Ch.30)                 | Containment-drill scenario                | SPECIFIED |
| Drift             | Verify drift detection and governed repair approval work end-to-end (Ch.32)    | Deliberate-drift detection test           | SPECIFIED |
| Game-day          | Verify combined failure-mode resilience under a controlled exercise            | Game-day exercise report                  | SPECIFIED |

34.2 Test-Status Integrity (directive item 65)

This chapter distinguishes TEST SPECIFIED, IMPLEMENTATION EVIDENCE AVAILABLE, TEST EXECUTED, TEST PASSED, GAME-DAY/RECOVERY VALIDATED and HUMAN/PRODUCTION ACCEPTANCE as six distinct states; every test class above is SPECIFIED only. Game-day success ≠ proof of every production failure mode — a passed exercise validates the exercised scenario, not every possible failure.

34.R RACI / Control Ownership

Table 42. Chapter 34 RACI / control ownership.

| Function                        | Accountable (NCIE-008)                       | Responsible (Operations)               | Consulted               | Informed                   |
|---------------------------------|----------------------------------------------|----------------------------------------|-------------------------|----------------------------|
| Confirm test acceptance/cadence | Accountable Authority Class (to be assigned) | Operations Test Owner (to be assigned) | NCIE-009/014/016 owners | Assurance/Audit (NCIE-008) |

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-34-1: Confirm test acceptance/cadence.

Traceability: NCIE-016/009/014.

# Human Review Register & Open Operations Decisions

Consolidate every unresolved operations decision exposed across Chapters 1-34 and Chapter 36 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008–014 governance, security, persistence and domain-business layers, or the approved NCIE-015 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008–014 Human Review registers, plus operations-specific unresolved items from Chapters 1-34.

## 35.0 Salient Things to Note

Do not invent products, SLOs, RPO/RTO, on-call models, windows or runbook authority; precise blocking scope required.

### 35.1 Purpose

This register exists so that no unresolved operations decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-34 below was extracted programmatically from that chapter's own Human Review Focus section (§35.4). Chapter 35's own item (HR15-35-1) and Chapter 36's own item (HR15-36-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

### 35.2 Summary

Table 43. Human Review Register summary (mechanically computed).

| Classification                                | Count |
|-----------------------------------------------|-------|
| ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED | 8     |
| INSTITUTIONAL CONFIRMATION REQUIRED           | 17    |
| LEGAL/POLICY CONFIRMATION REQUIRED            | 1     |
| RESOLVED BY UPSTREAM BASELINE                 | 1     |
| SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED    | 9     |
| Total                                         | 36    |
| — Blocking                                    | 32    |
| — Non-Blocking                                | 4     |

### 35.3 Complete Register

Table 44. Complete Human Review Register.

| Review ID | Source Chapter | Question                                                                          | Classification                                 | Blocking Scope                                                                    | Upstream Cross-Reference  | Status |
|-----------|----------------|-----------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------|--------|
| HR15-1-1  | Chapter 1      | Confirm the NCIE-015 formal title reconciliation (§0.1 — corrected to 'DevSecOps, | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking) | Confirm the NCIE-015 formal title reconciliation (§0.1 — corrected to 'DevSecOps, | N/A — distinct operations | Open   |

|          |           |                                                                                                                                                                                           |                                                              |                                                                                                                                                                                           |                                    |      |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|
|          |           | Infrastructure & Deployment Specification' against Master Controlled Register v2.0), acceptance/sign-off authority, and operations ownership (skeletal Preliminary Human Review Theme 1). |                                                              | Infrastructure & Deployment Specification' against Master Controlled Register v2.0), acceptance/sign-off authority, and operations ownership (skeletal Preliminary Human Review Theme 1). | decision                           |      |
| HR15-2-1 | Chapter 2 | Approve operational doctrine.                                                                                                                                                             | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Approve operational doctrine.                                                                                                                                                             | N/A — distinct operations decision | Open |
| HR15-3-1 | Chapter 3 | Confirm environment topology/data use (Table 5).                                                                                                                                          | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm environment topology/data use (Table 5).                                                                                                                                          | N/A — distinct operations decision | Open |
| HR15-4-1 | Chapter 4 | Confirm IaC/config technology and approval (no product mandated by this document per directive item 9).                                                                                   | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)     | Confirm IaC/config technology and approval (no product mandated by this document per directive item 9).                                                                                   | N/A — distinct operations decision | Open |
| HR15-5-1 | Chapter 5 | Confirm source-control model; does not block the review-before-merge baseline already defined in the interim.                                                                             | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking) | Non-blocking — does not gate chapter-level operational activity                                                                                                                           | N/A — distinct operations decision | Open |
| HR15-6-1 | Chapter 6 | Confirm build/artifact technology and gates.                                                                                                                                              | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm build/artifact technology and gates.                                                                                                                                              | N/A — distinct operations decision | Open |
| HR15-7-1 | Chapter 7 | Confirm mandatory gates/exceptions.                                                                                                                                                       | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Confirm mandatory gates/exceptions.                                                                                                                                                       | N/A — distinct operations decision | Open |
| HR15-8-1 | Chapter 8 | Confirm release authority/change classes (Table 11).                                                                                                                                      | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Confirm release authority/change classes (Table 11).                                                                                                                                      | N/A — distinct operations decision | Open |
| HR15-9-1 | Chapter 9 | Confirm permitted strategies; does not block the compatibility/rollback discipline already                                                                                                | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking) | Non-blocking — does not gate chapter-level operational activity                                                                                                                           | N/A — distinct operations decision | Open |

|           |            |                                                                                                                      |                                                              |                                                                            |                                    |      |
|-----------|------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------|------|
|           |            | defined in the interim.                                                                                              |                                                              |                                                                            |                                    |      |
| HR15-10-1 | Chapter 10 | Confirm runtime technology/classes (no product mandated by this document).                                           | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)     | Confirm runtime technology/classes (no product mandated by this document). | N/A — distinct operations decision | Open |
| HR15-11-1 | Chapter 11 | Confirm sandbox/runtime and generated-code activation.                                                               | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm sandbox/runtime and generated-code activation.                     | N/A — distinct operations decision | Open |
| HR15-12-1 | Chapter 12 | Confirm topology/egress classes.                                                                                     | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm topology/egress classes.                                           | N/A — distinct operations decision | Open |
| HR15-13-1 | Chapter 13 | Confirm privilege/on-call emergency model.                                                                           | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm privilege/on-call emergency model.                                 | N/A — distinct operations decision | Open |
| HR15-14-1 | Chapter 14 | Confirm custody/rotation policy.                                                                                     | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)        | Confirm custody/rotation policy.                                           | N/A — distinct operations decision | Open |
| HR15-15-1 | Chapter 15 | Confirm ownership/maintenance windows.                                                                               | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Confirm ownership/maintenance windows.                                     | N/A — distinct operations decision | Open |
| HR15-16-1 | Chapter 16 | Confirm platform ownership/thresholds.                                                                               | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)     | Confirm platform ownership/thresholds.                                     | N/A — distinct operations decision | Open |
| HR15-17-1 | Chapter 17 | Confirm service/on-call ownership.                                                                                   | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Confirm service/on-call ownership.                                         | N/A — distinct operations decision | Open |
| HR15-18-1 | Chapter 18 | Confirm module support ownership.                                                                                    | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)               | Confirm module support ownership.                                          | N/A — distinct operations decision | Open |
| HR15-19-1 | Chapter 19 | Confirm frontend topology; does not block the cache-invalidation/rollback discipline already defined in the interim. | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking) | Non-blocking — does not gate chapter-level operational activity            | N/A — distinct operations decision | Open |

|           |            |                                                                                           |                                                          |                                                                                           |                                    |      |
|-----------|------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------|------|
| HR15-20-1 | Chapter 20 | Confirm observability platform/retention (Table 24).                                      | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking) | Confirm observability platform/retention (Table 24).                                      | N/A — distinct operations decision | Open |
| HR15-21-1 | Chapter 21 | Confirm SLO catalogue/targets (no numeric value invented by this document).               | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm SLO catalogue/targets (no numeric value invented by this document).               | N/A — distinct operations decision | Open |
| HR15-22-1 | Chapter 22 | Confirm on-call/thresholds.                                                               | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm on-call/thresholds.                                                               | N/A — distinct operations decision | Open |
| HR15-23-1 | Chapter 23 | Confirm incident authority/severity (Table 28).                                           | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm incident authority/severity (Table 28).                                           | N/A — distinct operations decision | Open |
| HR15-24-1 | Chapter 24 | Confirm runbook ownership/automation (Table 30).                                          | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm runbook ownership/automation (Table 30).                                          | N/A — distinct operations decision | Open |
| HR15-25-1 | Chapter 25 | Confirm classes/windows/emergency authority.                                              | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm classes/windows/emergency authority.                                              | N/A — distinct operations decision | Open |
| HR15-26-1 | Chapter 26 | Confirm targets/limits (no numeric value invented by this document).                      | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking) | Confirm targets/limits (no numeric value invented by this document).                      | N/A — distinct operations decision | Open |
| HR15-27-1 | Chapter 27 | Confirm schedule/RPO/RTO/authority (no numeric value invented by this document).          | INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)           | Confirm schedule/RPO/RTO/authority (no numeric value invented by this document).          | N/A — distinct operations decision | Open |
| HR15-28-1 | Chapter 28 | Confirm DR authority/topology/cadence.                                                    | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)    | Confirm DR authority/topology/cadence.                                                    | N/A — distinct operations decision | Open |
| HR15-29-1 | Chapter 29 | Confirm remediation targets/risk interface (no deadline value invented by this document). | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)    | Confirm remediation targets/risk interface (no deadline value invented by this document). | N/A — distinct operations decision | Open |
| HR15-30-1 | Chapter 30 | Confirm SecOps ownership/escalation.                                                      | SECURITY/SOVEREIGNTY CONFIRMATION                        | Confirm SecOps ownership/escalation.                                                      | N/A — distinct operations decision | Open |

|           |            |                                                                                                                                                                                                                                                                         | REQUIRED (Blocking)                                  |                                                                                                                                                                                                    | ns<br>decision                     |                               |
|-----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|
| HR15-31-1 | Chapter 31 | Confirm record retention/access (no retention period invented by this document).                                                                                                                                                                                        | LEGAL/POLICY<br>CONFIRMATION<br>REQUIRED (Blocking)  | Confirm record retention/access (no retention period invented by this document).                                                                                                                   | N/A — distinct operations decision | Open                          |
| HR15-32-1 | Chapter 32 | Confirm auto-remediation boundaries.                                                                                                                                                                                                                                    | INSTITUTIONAL<br>CONFIRMATION<br>REQUIRED (Blocking) | Confirm auto-remediation boundaries.                                                                                                                                                               | N/A — distinct operations decision | Open                          |
| HR15-33-1 | Chapter 33 | Confirm go-live authority/evidence.                                                                                                                                                                                                                                     | INSTITUTIONAL<br>CONFIRMATION<br>REQUIRED (Blocking) | Confirm go-live authority/evidence.                                                                                                                                                                | N/A — distinct operations decision | Open                          |
| HR15-34-1 | Chapter 34 | Confirm test acceptance/cadence.                                                                                                                                                                                                                                        | INSTITUTIONAL<br>CONFIRMATION<br>REQUIRED (Blocking) | Confirm test acceptance/cadence.                                                                                                                                                                   | N/A — distinct operations decision | Open                          |
| HR15-35-1 | Chapter 35 | Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-015 v1.0 may be approved or may instead remain open under an interim design default.                                                                      | INSTITUTIONAL<br>CONFIRMATION<br>REQUIRED (Blocking) | Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-015 v1.0 may be approved or may instead remain open under an interim design default. | N/A — distinct operations decision | Open                          |
| HR15-36-1 | Chapter 36 | Approve the NCIE-015 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-015 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item. | RESOLVED BY UPSTREAM<br>BASELINE (Non-Blocking)      | Non-blocking — does not gate chapter-level operational activity                                                                                                                                    | N/A — distinct operations decision | Resolved by upstream baseline |

### 35.4 Mechanical Verification

34 of these 36 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-34 rather than assembled by hand. The remaining 2 items are HR15-35-1 (this chapter's own item) and HR15-36-1 (Chapter 36's own item), each defined once and rendered identically in both this table and their own chapter's

Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

### 35.5 Upstream Deduplication (Directive Item 64)

Every NCIE-015 Human Review item was individually checked against the existing NCIE-008–014 Human Review registers. No NCIE-015 item was found to restate an identical governance, security, persistence or domain-business question — each concerns a distinct operations decision (platform selection, numeric threshold, on-call/runbook authority, or ownership) that those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was therefore required for this edition.

### 35.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific platform selection, numeric threshold, on-call/runbook authority or ownership assignment — rather than implying the whole of NCIE-015 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's operational activity in the interim.

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR15-35-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-015 v1.0 may be approved or may instead remain open under an interim design default.

*Traceability: All chapters above (mechanically reconciled register); NCIE-008–014 Human Review registers.*

NCIE-015 — Chapter 36 Expansion

# Traceability, Acceptance & Downstream Handover

Close traceability to the approved NCIE-001-014 foundation and the approved NCIE-015 Skeletal v0.1; define acceptance criteria and the downstream handover to NCIE-016 and subsequent controlled documents.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Approved NCIE-001-014 foundation.

## 36.0 Salient Things to Note

Skeleton approval authorizes expansion only; approval does not prove deployment/readiness/continuity/test execution — §36.4's acceptance gate states this explicitly rather than leaving it implied.

## 36.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-015 Skeletal v0.1 itself. No chapter states an operations requirement without an identifiable source.

Table 45. Full traceability matrix.

| NCIE-015 Chapter | Primary Upstream Authority                | Relationship                                    |
|------------------|-------------------------------------------|-------------------------------------------------|
| Chapter 1        | NCIE-001-014, especially NCIE-004/009-014 | Mandate/authority chain/operations boundary     |
| Chapter 2        | NCIE-004/008/009                          | DevSecOps & operational doctrine                |
| Chapter 3        | NCIE-004/009/014                          | Environment model & promotion boundaries        |
| Chapter 4        | NCIE-004/008/009                          | IaC & configuration realization                 |
| Chapter 5        | NCIE-004/008                              | Source control, branching & change integration  |
| Chapter 6        | NCIE-004/009                              | Build, artifact & supply-chain pipeline         |
| Chapter 7        | NCIE-004/008/009/016                      | CI validation & pre-release gates               |
| Chapter 8        | NCIE-008/004                              | Release management & deployment authorization   |
| Chapter 9        | NCIE-004/011/014                          | Deployment strategies & rollout control         |
| Chapter 10       | NCIE-004/009                              | Runtime platform & workload scheduling          |
| Chapter 11       | NCIE-007/009/004                          | Agent runtime & sandbox infrastructure          |
| Chapter 12       | NCIE-009/004                              | Network, ingress, egress & service connectivity |
| Chapter 13       | NCIE-009/008                              | IAM, workload identity & privileged operations  |
| Chapter 14       | NCIE-009/004                              | Secrets, certificates & key operations          |
| Chapter 15       | NCIE-014                                  | Database & storage deployment operations        |
| Chapter 16       | NCIE-011                                  | API gateway, event bus & connector operations   |

|            |                                                         |                                                             |
|------------|---------------------------------------------------------|-------------------------------------------------------------|
| Chapter 17 | NCIE-005/007/009/011                                    | ARGUS, Model Gateway, Tool Gateway & VPF operations         |
| Chapter 18 | NCIE-013                                                | Functional module operations                                |
| Chapter 19 | NCIE-012/009/011                                        | UX/frontend delivery operations                             |
| Chapter 20 | NCIE-004/009/010                                        | Observability architecture & telemetry operations           |
| Chapter 21 | NCIE-002/004/013/014                                    | SLOs, indicators & error budgets                            |
| Chapter 22 | NCIE-004/008/012                                        | Alerting, on-call & operational escalation                  |
| Chapter 23 | NCIE-009/008/010                                        | Incident management & command                               |
| Chapter 24 | NCIE-009-014 handovers                                  | Runbooks, playbooks & operational procedures                |
| Chapter 25 | NCIE-008/009                                            | Change, maintenance & emergency operations                  |
| Chapter 26 | NCIE-002/004/014                                        | Capacity, scaling & performance operations                  |
| Chapter 27 | NCIE-014/009/010                                        | Backup, restore & data-protection operations                |
| Chapter 28 | NCIE-004/014/009                                        | Disaster recovery, failover & business continuity           |
| Chapter 29 | NCIE-009/008/004                                        | Vulnerability, patch & remediation operations               |
| Chapter 30 | NCIE-009/010                                            | Operational security monitoring & response integration      |
| Chapter 31 | NCIE-010/008/011                                        | Operational audit, Evidence & change provenance             |
| Chapter 32 | NCIE-004/008/014                                        | Operational reconciliation, drift & configuration integrity |
| Chapter 33 | NCIE-008/016                                            | Operational readiness, handover & go-live governance        |
| Chapter 34 | NCIE-016/009/014                                        | DevSecOps/operations testing & game-day validation          |
| Chapter 35 | All chapters above; NCIE-008-014 Human Review registers | Consolidated Human Review Register                          |
| Chapter 36 | Approved NCIE-001-014; NCIE-015 Skeletal v0.1           | Traceability/acceptance/handover                            |

### 36.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 36 sequential 'NCIE-015 — Chapter N Expansion' labels, N = 1..36, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 77). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 36 chapters (directive item 78), rerun independently against this final document rather than assumed from the build script.

### 36.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-015 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 76); none is omitted, and none is conceptually inverted.

36.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-015 v1.0 requires: (a) 36/36 chapter integrity confirmed; (b) 36/36 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 35 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority (including the formal-title reconciliation, §0.1, HR15-1-1); (e) the §34.1 test suite specified with no test falsely marked executed, passed, or game-day/recovery-validated; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any platform selection, numeric threshold, on-call model, runbook authority or ownership assignment proposed above. Operations approval under this chapter does not prove deployment success, recovery validity, capacity sufficiency or production readiness.

36.5 Downstream Handover Register

NCIE-015 does not preempt detailed downstream specification. The table below states what is handed over and what remains downstream.

Table 46. Downstream handover register.

| Downstream Document             | What NCIE-015 Hands Over                                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCIE-016 (detailed testing)     | The 10 DevSecOps/operations test classes specified in §34.1 (Table 41); detailed test cases, execution and game-day/recovery validation remain downstream in NCIE-016 — this document specifies the tests, not their execution. |
| Subsequent controlled documents | The complete set of open operations Human Review items (Chapter 35) as build-sequence and operational-readiness dependencies for any later controlled document that consumes NCIE-015's operational architecture.               |

36.6 Readiness and Final Disposition

NCIE-015 DevSecOps, Infrastructure & Deployment Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed platform selection, numeric threshold, on-call model, runbook authority and ownership assignment deferred in Chapters 1-34 remains a proposal pending the Chapter 35 Human Review Register; every §34.1 test remains SPECIFIED, not executed, passed, or game-day/recovery-validated, pending NCIE-016.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR15-36-1: Approve the NCIE-015 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-015 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-014 approved foundation; approved NCIE-015 Skeletal v0.1.